

Aggregate Functions (COUNT, SUM, AVG)

Step 1: Create the Table

None

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    employee_name VARCHAR(50),  
    department VARCHAR(30),  
    salary INT,  
    experience INT  
);
```

Step 2: Insert Sample Data

None

```
INSERT INTO employees VALUES  
(101, 'Rahul', 'HR', 45000, 2),  
(102, 'Anu', 'IT', 65000, 5),  
(103, 'John', 'Sales', 55000, 3),  
(104, 'Meera', 'IT', 70000, 6),  
(105, 'Kiran', 'Finance', 80000, 8),  
(106, 'Divya', 'Marketing', 50000, 4),  
(107, 'Arun', 'IT', 75000, 7),  
(108, 'Sneha', 'Sales', 60000, 5);
```

COUNT()

1. Count the total number of employees.

None

```
SELECT COUNT(*) AS total_employees  
FROM employees;
```

2. Count employees in the IT department.

None

```
SELECT COUNT(*) AS it_employees  
FROM employees  
WHERE department = 'IT';
```

3. Count employees whose salary is greater than 60,000.

None

```
SELECT COUNT(*) AS high_salary_employees  
FROM employees  
WHERE salary > 60000;
```

SUM()

4. Find the total salary paid to all employees.

None

```
SELECT SUM(salary) AS total_salary  
FROM employees;
```

5. Find the total salary of IT employees.

None

```
SELECT SUM(salary) AS total_it_salary  
FROM employees
```

```
WHERE department = 'IT';
```

6. Find the total years of experience of all employees.

None

```
SELECT SUM(experience) AS total_experience  
FROM employees;
```

AVG()

7. Find the average salary.

None

```
SELECT AVG(salary) AS average_salary  
FROM employees;
```

8. Find the average salary in the Sales department.

None

```
SELECT AVG(salary) AS average_sales_salary  
FROM employees  
WHERE department = 'Sales';
```

9. Find the average experience of employees.

None

```
SELECT AVG(experience) AS average_experience  
FROM employees;
```

Combined Aggregate Functions

10. Display the total number of employees, total salary, and average salary.

None

```
SELECT
    COUNT(*) AS total_employees,
    SUM(salary) AS total_salary,
    AVG(salary) AS average_salary
FROM employees;
```

MIN()

1. Find the minimum salary.

None

```
SELECT MIN(salary) AS minimum_salary
FROM employees;
```

2. Find the minimum experience.

None

```
SELECT MIN(experience) AS minimum_experience
FROM employees;
```

3. Find the minimum salary in the IT department.

None

```
SELECT MIN(salary) AS minimum_it_salary
FROM employees
WHERE department = 'IT';
```

4. Find the employee(s) earning the minimum salary.

None

```
SELECT *  
FROM employees  
WHERE salary = (  
    SELECT MIN(salary)  
    FROM employees  
);
```

MAX()

5. Find the maximum salary.

None

```
SELECT MAX(salary) AS maximum_salary  
FROM employees;
```

6. Find the maximum experience.

None

```
SELECT MAX(experience) AS maximum_experience  
FROM employees;
```

7. Find the maximum salary in the Sales department.

None

```
SELECT MAX(salary) AS maximum_sales_salary  
FROM employees  
WHERE department = 'Sales';
```

8. Find the employee(s) earning the highest salary.

None

```
SELECT *  
FROM employees  
WHERE salary = (  
    SELECT MAX(salary)  
    FROM employees  
);
```

MIN() and MAX() Together

9. Display the minimum and maximum salary.

None

```
SELECT  
    MIN(salary) AS minimum_salary,  
    MAX(salary) AS maximum_salary  
FROM employees;
```

10. Display the minimum and maximum experience.

None

```
SELECT  
    MIN(experience) AS minimum_experience,  
    MAX(experience) AS maximum_experience  
FROM employees;
```